E-Commerce Database Management System

Week 4: Order Management System

1. Orders and Order_Details Table Design

Week 4 turns the orders table from Task 1 into a complete order management system. An order can contain many products, and a product can appear in many orders. A relational table cannot store this directly, so a new order_details table holds one row for each product in an order.

Week 4 objectives

- Use the orders table and create the order_details table, with the Customer to Orders, Orders to Order_Details and Product to Order_Details relationships.
- Apply primary key, foreign key, NOT NULL, UNIQUE and CHECK constraints.
- Create an order, add several products to it, calculate the total amount, update the status and display the complete order.
- Update the quantity of an ordered product and delete cancelled orders.
- Generate customer order history and sales reports.

How Week 4 builds on earlier tasks. The orders table from Task 1 already has the Week 4 columns, and orders.customer_id already links each order to its customer. The new order_details table connects orders to products. Order details are added for the 10 orders from Task 1, and five new multi-product orders are created. Payments for the new orders are added in Week 5.

Design rules used in this week

- The price in order_details is the price at the time of the order. A later change to the product price does not alter old orders. For example, order 1 keeps 17999.00 although the current price of that product is 16999.00.
- Sales reports ignore Cancelled orders. The order history report shows every order.
- Stock is not reduced when an order is placed, so the Week 3 inventory figures stay the same.

1.1 Orders (table: orders)

Attribute	Data type	Constraint	Description
order_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the order
customer_id	INT	NOT NULL, FOREIGN KEY references customer(customer_id)	Customer who placed the order

Attribute	Data type	Constraint	Description
order_date	DATE	NOT NULL	Date the order was placed
total_amount	DECIMAL(10,2)	NOT NULL, CHECK (total_amount >= 0)	Total value of the order
order_status	ENUM	NOT NULL, DEFAULT 'Pending'	Pending, Shipped, Delivered or Cancelled

1.2 Order_Details (table: order_details)

Attribute	Data type	Constraint	Description
order_detail_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID of the order line
order_id	INT	NOT NULL, FOREIGN KEY references orders(order_id), ON DELETE CASCADE	Order the line belongs to
product_id	INT	NOT NULL, FOREIGN KEY references product(product_id)	Product that was ordered
quantity	INT	NOT NULL, CHECK (quantity > 0)	Number of units ordered
price	DECIMAL(10,2)	NOT NULL, CHECK (price > 0)	Price per unit when the order was placed

The pair (order_id, product_id) is also UNIQUE, so a product appears only once in an order. To order more units, the quantity is increased.

1.3 Order Status Values

Status	Meaning
Pending	The order is placed and not yet shipped
Shipped	The order has been sent to the customer
Delivered	The customer has received the order
Cancelled	The order is cancelled and is left out of the sales reports

1.4 How the Order Total Is Calculated

The total of an order is the sum of quantity multiplied by price over all its rows in order_details. For order 11 the total is 1 x 1799.00 + 2 x 1299.00 + 1 x 999.00 = 5396.00.

1.5 Constraint Summary

Constraint	Purpose	Where it is used
PRIMARY KEY	Identifies each row uniquely	orders.order_id, order_details.order_detail_id
FOREIGN KEY	Links a row to a row in another table	orders.customer_id to customer, order_details.order_id to orders, order_details.product_id to product
NOT NULL	Makes a value compulsory	Every column of order_details, customer_id, order_date and total_amount in orders
UNIQUE	Prevents duplicate values	The pair (order_id, product_id) in order_details
CHECK	Rejects invalid values	quantity > 0, price > 0, total_amount >= 0
DEFAULT	Supplies a value when none is given	order_status ('Pending')

1.6 Relationships

Relationship	Type	Linked by	Explanation
Customer to Orders	One-to-Many	orders.customer_id	One customer can place many orders. Each order belongs to one customer.
Orders to Order_Details	One-to-Many	order_details.order_id	One order contains many order lines. Each line belongs to one order.
Product to Order_Details	One-to-Many	order_details.product_id	One product can appear in many order lines. Each line refers to one product.
Orders and Product	Many-to-Many	Order_Details table	An order has many products and a product is in many orders. Order_Details turns this into two one-to-many links.

Referential integrity rules

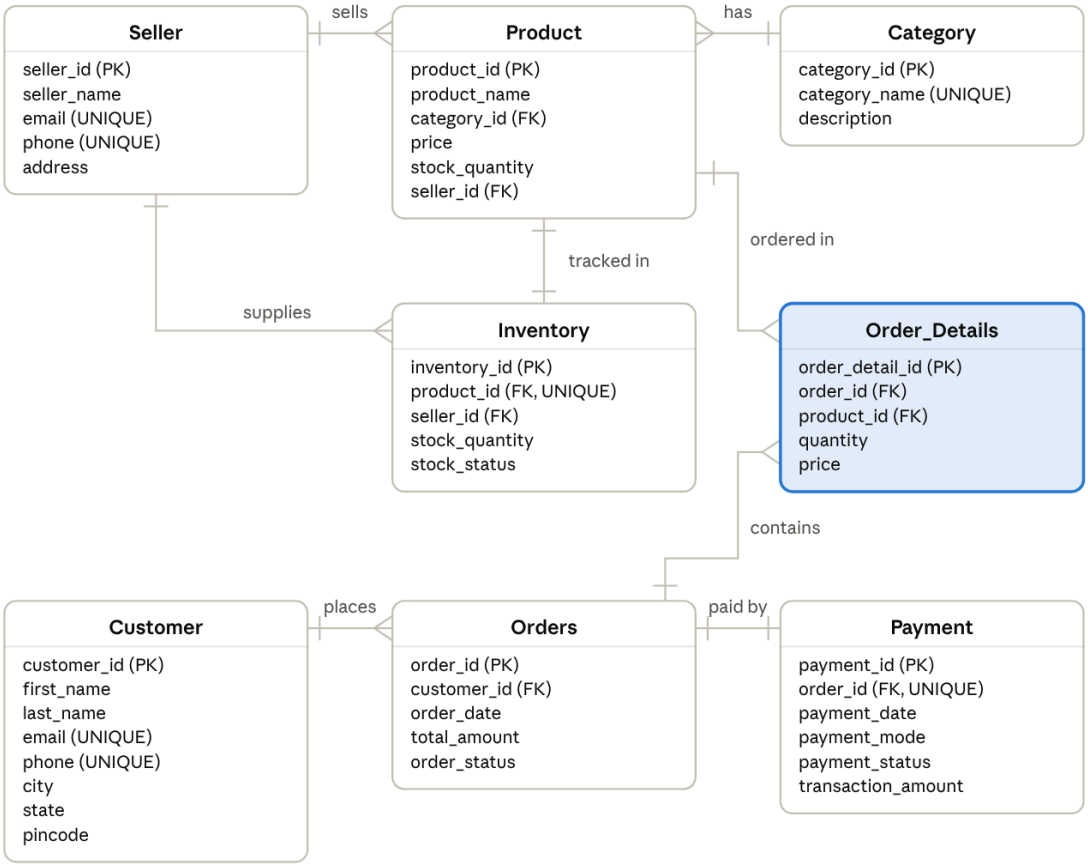
- An order cannot be created for a customer who does not exist.
- Deleting an order also deletes its order lines (ON DELETE CASCADE).
- A product that appears in an order cannot be deleted (MySQL error 1451), so order history is protected.

- The same product cannot be added twice to one order, and a quantity must be greater than 0.

2. ER Diagram

The updated ER diagram adds the order_details table. It replaces the dotted many-to-many link between Orders and Product that was shown in the earlier diagrams. The highlighted table is new in Week 4, and the crow's foot marks the many side of a relationship.

Order_Details is new in Week 4: it links each order to its products



PK = primary key, FK = foreign key. A bar means one, a three-pronged foot means many.
The highlighted table is new in Week 4. Each product appears once per order; quantity holds the count.

Updated ER diagram · 8 entities, 9 relationships

Each order contains many order lines, and each line refers to one product. A customer places many orders, and each order has at most one payment. The relationships from Tasks 1 to 3 are unchanged.

3. Sample Customer, Order and Order_Details Data

3.1 Customer data

Nine customers exist from Task 1. Customer 10 was removed by the DELETE example in Task 1, and the city of customer 1 was changed to Chennai.

Customer_ID	Name	Email	City	State
1	Arun Kumar	arun@mail.com	Chennai	Tamil Nadu
2	Priya Sharma	priya@mail.com	Chennai	Tamil Nadu
3	Rahul Verma	rahul@mail.com	Mumbai	Maharashtra
4	Sneha Reddy	sneha@mail.com	Hyderabad	Telangana
5	Karthik Raja	karthik@mail.com	Madurai	Tamil Nadu
6	Divya Nair	divya@mail.com	Kochi	Kerala
7	Vikram Singh	vikram@mail.com	Delhi	Delhi
8	Meena Iyer	meena@mail.com	Coimbatore	Tamil Nadu
9	Suresh Babu	suresh@mail.com	Bengaluru	Karnataka

3.2 Order data

Orders 1 to 10 come from Task 1. Orders 11 to 15 are new. Their totals are calculated from the order lines with one UPDATE statement. The statuses below are the ones at the start of Week 4.

Order_ID	Customer	Order_Date	Total_Amount	Order_Status
1	Arun Kumar	2026-10-01	17999.00	Delivered
2	Priya Sharma	2026-10-02	998.00	Delivered
3	Rahul Verma	2026-10-03	1899.00	Shipped
4	Sneha Reddy	2026-10-04	750.00	Shipped
5	Karthik Raja	2026-10-05	1299.00	Shipped
6	Divya Nair	2026-10-06	1499.00	Pending
7	Vikram Singh	2026-10-07	3299.00	Delivered
8	Meena Iyer	2026-10-08	398.00	Cancelled
9	Suresh Babu	2026-10-09	799.00	Pending

Order_ID	Customer	Order_Date	Total_Amount	Order_Status
10	Arun Kumar	2026-10-10	700.00	Pending
11	Rahul Verma	2026-10-11	5396.00	Delivered
12	Arun Kumar	2026-10-12	3697.00	Shipped
13	Priya Sharma	2026-10-13	1996.00	Pending
14	Vikram Singh	2026-10-14	3998.00	Delivered
15	Sneha Reddy	2026-10-15	897.00	Cancelled

3.3 Order_Details data

Orders 1 to 10 have one product each. Orders 11 to 15 have two or three products each.

Order_Detail_ID	Order_ID	Product	Quantity	Price	Line total
1	1	Redmi Note 13 5G	1	17999.00	17999.00
2	2	Men Cotton T-Shirt	2	499.00	998.00
3	3	Non-stick Cookware Set	1	1899.00	1899.00
4	4	Wings of Fire (Book)	3	250.00	750.00
5	5	Kashmir Willow Cricket Bat	1	1299.00	1299.00
6	6	Wireless Headphones	1	1499.00	1499.00
7	7	Mixer Grinder 750W	1	3299.00	3299.00
8	8	Face Wash 100ml	2	199.00	398.00
9	9	Building Blocks Set	1	799.00	799.00
10	10	Organic Honey 500g	2	350.00	700.00
11	11	Bluetooth Speaker	1	1799.00	1799.00
12	11	Power Bank 20000mAh	2	1299.00	2598.00
13	11	USB-C Fast Charger 65W	1	999.00	999.00
14	12	Running Shoes	1	2299.00	2299.00
15	12	Women Kurti	2	699.00	1398.00
16	13	Steel Water Bottle 1L	2	399.00	798.00

Order_Detail_ID	Order_ID	Product	Quantity	Price	Line total
17	13	Atomic Habits (Book)	1	499.00	499.00
18	13	Yoga Mat	1	699.00	699.00
19	14	Smart Watch	1	2499.00	2499.00
20	14	Wireless Headphones	1	1499.00	1499.00
21	15	Moisturizer 200ml	2	349.00	698.00
22	15	Face Wash 100ml	1	199.00	199.00

3.4 Changes made by the order operations

Operation	Change
New order	Order 16 is created for Divya Nair on 2026-10-16 with Mixer Grinder 750W (1 x 3299.00) and Non-stick Cookware Set (1 x 1899.00). The total is calculated as 5198.00 and the status changes from Pending to Shipped.
Status updates	Order 13 changes from Pending to Shipped. Orders 12 and 5 change from Shipped to Delivered.
Quantity update	In order 13 the quantity of Steel Water Bottle 1L changes from 2 to 3, and the order total changes from 1996.00 to 2395.00.
Delete cancelled order	Order 15 is deleted together with its two order lines. Order 8 is also cancelled but is kept, because its failed payment record is used in Week 5.

The analysis report below uses the data after these changes.

4. Customer Order History and Purchase Analysis Report

This report presents the results of the order history and sales queries in the SQL script, followed by the main observations. The sales figures leave out Cancelled orders.

4.1 Customer order history

The history shows every order, including the cancelled one.

Customer	Order_ID	Order_Date	Order_Status	Total_Amount
Arun Kumar	1	2026-10-01	Delivered	17999.00

Customer	Order_ID	Order_Date	Order_Status	Total_Amount
Arun Kumar	10	2026-10-10	Pending	700.00
Arun Kumar	12	2026-10-12	Delivered	3697.00
Priya Sharma	2	2026-10-02	Delivered	998.00
Priya Sharma	13	2026-10-13	Shipped	2395.00
Rahul Verma	3	2026-10-03	Shipped	1899.00
Rahul Verma	11	2026-10-11	Delivered	5396.00
Sneha Reddy	4	2026-10-04	Shipped	750.00
Karthik Raja	5	2026-10-05	Delivered	1299.00
Divya Nair	6	2026-10-06	Pending	1499.00
Divya Nair	16	2026-10-16	Shipped	5198.00
Vikram Singh	7	2026-10-07	Delivered	3299.00
Vikram Singh	14	2026-10-14	Delivered	3998.00
Meena Iyer	8	2026-10-08	Cancelled	398.00
Suresh Babu	9	2026-10-09	Pending	799.00

4.2 Orders and sales by status

Order_Status	Total_Orders	Total_Amount
Pending	3	2998.00
Shipped	4	10242.00
Delivered	7	36686.00
Cancelled	1	398.00

4.3 Product-wise order report

Product	Order_ID	Customer	Quantity	Price	Line total
Atomic Habits (Book)	13	Priya Sharma	1	499.00	499.00
Bluetooth Speaker	11	Rahul Verma	1	1799.00	1799.00

Product	Order_ID	Customer	Quantity	Price	Line total
Building Blocks Set	9	Suresh Babu	1	799.00	799.00
Kashmir Willow Cricket Bat	5	Karthik Raja	1	1299.00	1299.00
Men Cotton T-Shirt	2	Priya Sharma	2	499.00	998.00
Mixer Grinder 750W	7	Vikram Singh	1	3299.00	3299.00
Mixer Grinder 750W	16	Divya Nair	1	3299.00	3299.00
Non-stick Cookware Set	3	Rahul Verma	1	1899.00	1899.00
Non-stick Cookware Set	16	Divya Nair	1	1899.00	1899.00
Organic Honey 500g	10	Arun Kumar	2	350.00	700.00
Power Bank 20000mAh	11	Rahul Verma	2	1299.00	2598.00
Redmi Note 13 5G	1	Arun Kumar	1	17999.00	17999.00
Running Shoes	12	Arun Kumar	1	2299.00	2299.00
Smart Watch	14	Vikram Singh	1	2499.00	2499.00
Steel Water Bottle 1L	13	Priya Sharma	3	399.00	1197.00
USB-C Fast Charger 65W	11	Rahul Verma	1	999.00	999.00
Wings of Fire (Book)	4	Sneha Reddy	3	250.00	750.00
Wireless Headphones	6	Divya Nair	1	1499.00	1499.00
Wireless Headphones	14	Vikram Singh	1	1499.00	1499.00
Women Kurti	12	Arun Kumar	2	699.00	1398.00
Yoga Mat	13	Priya Sharma	1	699.00	699.00

4.4 Number of times each product was ordered

Product	Times_Ordered	Total_Quantity	Total_Revenue
Mixer Grinder 750W	2	2	6598.00
Non-stick Cookware Set	2	2	3798.00
Wireless Headphones	2	2	2998.00
Steel Water Bottle 1L	1	3	1197.00

Product	Times_Ordered	Total_Quantity	Total_Revenue
Wings of Fire (Book)	1	3	750.00
Men Cotton T-Shirt	1	2	998.00
Organic Honey 500g	1	2	700.00
Power Bank 20000mAh	1	2	2598.00
Women Kurti	1	2	1398.00
Atomic Habits (Book)	1	1	499.00
Bluetooth Speaker	1	1	1799.00
Building Blocks Set	1	1	799.00
Kashmir Willow Cricket Bat	1	1	1299.00
Redmi Note 13 5G	1	1	17999.00
Running Shoes	1	1	2299.00
Smart Watch	1	1	2499.00
USB-C Fast Charger 65W	1	1	999.00
Yoga Mat	1	1	699.00

Eight products have never been ordered in a valid order: Face Wash 100ml, Moisturizer 200ml, Denim Jacket, Leather Wallet, Almonds 500g, Green Tea 100 Bags, Cotton Bedsheet and Cricket Ball Pack. Face Wash 100ml appears only in the cancelled order 8, and Moisturizer 200ml appeared only in the deleted order 15.

4.5 Total spending by each customer

Customer	Total_Orders	Total_Spent
Arun Kumar	3	22396.00
Vikram Singh	2	7297.00
Rahul Verma	2	7295.00
Divya Nair	2	6697.00
Priya Sharma	2	3393.00
Karthik Raja	1	1299.00
Suresh Babu	1	799.00

Customer	Total_Orders	Total_Spent
Sneha Reddy	1	750.00
Meena Iyer	0	0.00

The query uses LEFT JOIN, so Meena Iyer still appears with 0 because her only order was cancelled. The customer with the maximum number of orders is **Arun Kumar**, with 3 orders.

4.6 Sales figures

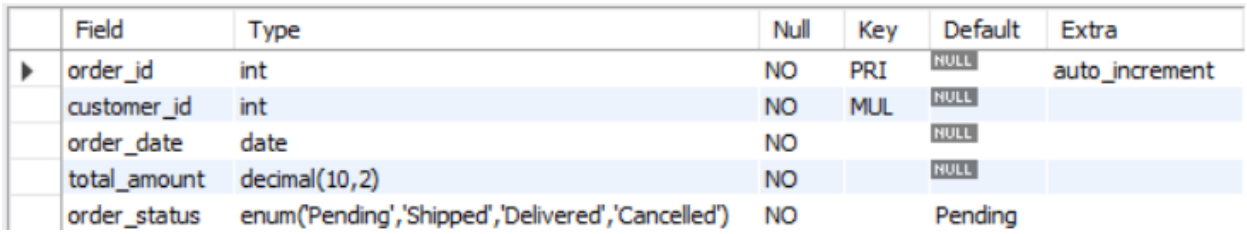
Measure	Result
Total quantity of products sold	29 units
Number of valid orders (not cancelled)	14
Total sales amount	49926.00
Average order value	3566.14

4.7 Key observations

- Arun Kumar is the top customer. He placed 3 of the 14 valid orders and spent 22396.00, about 45% of total sales, mostly because of the Redmi Note 13 5G order.
- Delivered orders make up 36686.00, about 73.5% of sales. Shipped orders add 10242.00 (20.5%) and pending orders 2998.00 (6.0%).
- The average order value of 3566.14 is pulled up by the 17999.00 Redmi order. The average of the other 13 orders is about 2455.92.
- The five multi-product orders (11, 12, 13, 14 and 16) bring in 20684.00, about 41% of sales, from only 5 of the 14 orders.
- Mixer Grinder 750W, Non-stick Cookware Set and Wireless Headphones are the only products ordered twice.
- Eight products have not been ordered yet, which shows where promotion or stock review is needed. Meena Iyer has no valid order because her only order was cancelled.

5. Query Output Screenshots

1. DESCRIBE orders



Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	auto_increment
customer_id	int	NO	MUL	NULL	
order_date	date	NO		NULL	
total_amount	decimal(10,2)	NO		NULL	
order_status	enum('Pending','Shipped','Delivered','Cancelled')	NO		Pending	

2. DESCRIBE order_details

Field	Type	Null	Key	Default	Extra
order_detail_id	int	NO	PRI	NULL	auto_increment
order_id	int	NO	MUL	NULL	
product_id	int	NO	MUL	NULL	
quantity	int	NO		NULL	
price	decimal(10,2)	NO		NULL	

3. SELECT * FROM order_details

order_detail_id	order_id	product_id	quantity	price
1	1	1	1	17999.00
2	2	2	2	499.00
3	3	3	1	1899.00
4	4	4	3	250.00
5	5	5	1	1299.00
6	6	6	1	1499.00
7	7	7	1	3299.00
8	8	8	2	199.00
9	9	9	1	799.00
10	10	10	2	350.00
11	11	11	1	1799.00
12	11	13	2	1299.00
13	11	14	1	999.00
14	12	16	1	2299.00
15	12	15	2	699.00
16	13	17	3	399.00
17	13	18	1	499.00
18	13	19	1	699.00
19	14	12	1	2499.00
20	14	6	1	1499.00
23	16	7	1	3299.00
24	16	3	1	1899.00
NULL	NULL	NULL	NULL	NULL

4. SELECT order_id, customer_id, order_date, total_amount, order_status FROM orders

order_id	customer_id	order_date	total_amount	order_status
1	1	2026-10-01	17999.00	Delivered
2	2	2026-10-02	998.00	Delivered
3	3	2026-10-03	1899.00	Shipped
4	4	2026-10-04	750.00	Shipped
5	5	2026-10-05	1299.00	Delivered
6	6	2026-10-06	1499.00	Pending
7	7	2026-10-07	3299.00	Delivered
8	8	2026-10-08	398.00	Cancelled
9	9	2026-10-09	799.00	Pending
10	1	2026-10-10	700.00	Pending
11	3	2026-10-11	5396.00	Delivered
12	1	2026-10-12	3697.00	Delivered
13	2	2026-10-13	2395.00	Shipped
14	7	2026-10-14	3998.00	Delivered
16	6	2026-10-16	5198.00	Shipped
NULL	NULL	NULL	NULL	NULL

5. Calculated total of the new order

order_id	calculated_total
16	5198.00

6. UPDATE orders SET order_status = 'Shipped' WHERE order_id = 13;

185 02:19:47 UPDATE orders SET order_status = 'Shipped' WHERE order_id = 13

UPDATE orders SET order_status = 'Delivered' WHERE order_id = 12;

186 02:19:47 UPDATE orders SET order_status = 'Delivered' WHERE order_id = 12

UPDATE orders SET order_status = 'Delivered' WHERE order_id = 5;

187 02:19:47 UPDATE orders SET order_status = 'Delivered' WHERE order_id = 5

SELECT order_id, order_status FROM orders WHERE order_id IN (5, 12, 13);

order_id	order_status
5	Delivered
12	Delivered
13	Shipped
NULL	NULL

7. UPDATE order_details SET quantity = 3 WHERE order_id = 13 AND product_id = 17;

```
189  02:19:47  UPDATE order_details SET quantity = 3 WHERE order_id = 13 AND product_id = 17
```

UPDATE orders

SET total_amount = (SELECT SUM(quantity * price) FROM order_details WHERE order_id = 13)

WHERE order_id = 13;

```
190  02:19:47  UPDATE orders SET total_amount = (SELECT SUM(quantity * price) FROM order_details WHERE order_id = ...
```

SELECT od.order_id, p.product_name, od.quantity, od.price, o.total_amount

FROM order_details od

JOIN orders o ON od.order_id = o.order_id

JOIN product p ON od.product_id = p.product_id

WHERE od.order_id = 13;

order_id	product_name	quantity	price	total_amount
13	Steel Water Bottle 1L	3	399.00	2395.00
13	Atomic Habits (Book)	1	499.00	2395.00
13	Yoga Mat	1	699.00	2395.00

8. Cancelled orders and the lines of order 15

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194  02:19:47  DELETE FROM orders WHERE order_id = 15 AND order_status = 'Cancelled'
```

9. Complete order information

order_id	order_date	order_status	customer_name	city	product_name	quantity	price	line_total	total_amount
1	2026-10-01	Delivered	Arun Kumar	Chennai	Redmi Note 13 5G	1	17999.00	17999.00	17999.00
2	2026-10-02	Delivered	Priya Sharma	Chennai	Men Cotton T-Shirt	2	499.00	998.00	998.00
3	2026-10-03	Shipped	Rahul Verma	Mumbai	Non-stick Cookware Set	1	1899.00	1899.00	1899.00
4	2026-10-04	Shipped	Sneha Reddy	Hyderabad	Wings of Fire (Book)	3	250.00	750.00	750.00
5	2026-10-05	Delivered	Karthik Raja	Madurai	Kashmir Willow Cricket Bat	1	1299.00	1299.00	1299.00
6	2026-10-06	Pending	Divya Nair	Kochi	Wireless Headphones	1	1499.00	1499.00	1499.00
7	2026-10-07	Delivered	Vikram Singh	Delhi	Mixer Grinder 750W	1	3299.00	3299.00	3299.00
8	2026-10-08	Cancelled	Meena Iyer	Coimbatore	Face Wash 100ml	2	199.00	398.00	398.00
9	2026-10-09	Pending	Suresh Babu	Bengaluru	Building Blocks Set	1	799.00	799.00	799.00
10	2026-10-10	Pending	Arun Kumar	Chennai	Organic Honey 500g	2	350.00	700.00	700.00
11	2026-10-11	Delivered	Rahul Verma	Mumbai	Bluetooth Speaker	1	1799.00	1799.00	5396.00
11	2026-10-11	Delivered	Rahul Verma	Mumbai	Power Bank 20000mAh	2	1299.00	2598.00	5396.00
11	2026-10-11	Delivered	Rahul Verma	Mumbai	USB-C Fast Charger 65W	1	999.00	999.00	5396.00
12	2026-10-12	Delivered	Arun Kumar	Chennai	Running Shoes	1	2299.00	2299.00	3697.00
12	2026-10-12	Delivered	Arun Kumar	Chennai	Women Kurti	2	699.00	1398.00	3697.00
13	2026-10-13	Shipped	Priya Sharma	Chennai	Atomic Habits (Book)	1	499.00	499.00	2395.00
13	2026-10-13	Shipped	Priya Sharma	Chennai	Steel Water Bottle 1L	3	399.00	1197.00	2395.00
13	2026-10-13	Shipped	Priya Sharma	Chennai	Yoga Mat	1	699.00	699.00	2395.00
14	2026-10-14	Delivered	Vikram Singh	Delhi	Smart Watch	1	2499.00	2499.00	3998.00
14	2026-10-14	Delivered	Vikram Singh	Delhi	Wireless Headphones	1	1499.00	1499.00	3998.00
16	2026-10-16	Shipped	Divya Nair	Kochi	Mixer Grinder 750W	1	3299.00	3299.00	5198.00
16	2026-10-16	Shipped	Divya Nair	Kochi	Non-stick Cookware Set	1	1899.00	1899.00	5198.00

10. Customer order history

customer_id	customer_name	order_id	order_date	order_status	total_amount
1	Arun Kumar	1	2026-10-01	Delivered	17999.00
1	Arun Kumar	10	2026-10-10	Pending	700.00
1	Arun Kumar	12	2026-10-12	Delivered	3697.00
2	Priya Sharma	2	2026-10-02	Delivered	998.00
2	Priya Sharma	13	2026-10-13	Shipped	2395.00
3	Rahul Verma	3	2026-10-03	Shipped	1899.00
3	Rahul Verma	11	2026-10-11	Delivered	5396.00
4	Sneha Reddy	4	2026-10-04	Shipped	750.00
5	Karthik Raja	5	2026-10-05	Delivered	1299.00
6	Divya Nair	6	2026-10-06	Pending	1499.00
6	Divya Nair	16	2026-10-16	Shipped	5198.00
7	Vikram Singh	7	2026-10-07	Delivered	3299.00
7	Vikram Singh	14	2026-10-14	Delivered	3998.00
8	Meena Iyer	8	2026-10-08	Cancelled	398.00
9	Suresh Babu	9	2026-10-09	Pending	799.00

11. Product-wise order report

product_name	order_id	customer_name	quantity	price	line_total
Atomic Habits (Book)	13	Priya Sharma	1	499.00	499.00
Bluetooth Speaker	11	Rahul Verma	1	1799.00	1799.00
Building Blocks Set	9	Suresh Babu	1	799.00	799.00
Kashmir Willow Cricket Bat	5	Karthik Raja	1	1299.00	1299.00
Men Cotton T-Shirt	2	Priya Sharma	2	499.00	998.00
Mixer Grinder 750W	7	Vikram Singh	1	3299.00	3299.00
Mixer Grinder 750W	16	Divya Nair	1	3299.00	3299.00
Non-stick Cookware Set	3	Rahul Verma	1	1899.00	1899.00
Non-stick Cookware Set	16	Divya Nair	1	1899.00	1899.00
Organic Honey 500g	10	Arun Kumar	2	350.00	700.00
Power Bank 20000mAh	11	Rahul Verma	2	1299.00	2598.00
Redmi Note 13 5G	1	Arun Kumar	1	17999.00	17999.00
Running Shoes	12	Arun Kumar	1	2299.00	2299.00
Smart Watch	14	Vikram Singh	1	2499.00	2499.00
Steel Water Bottle 1L	13	Priya Sharma	3	399.00	1197.00
USB-C Fast Charger 65W	11	Rahul Verma	1	999.00	999.00
Wings of Fire (Book)	4	Sneha Reddy	3	250.00	750.00
Wireless Headphones	6	Divya Nair	1	1499.00	1499.00
Wireless Headphones	14	Vikram Singh	1	1499.00	1499.00
Women Kurti	12	Arun Kumar	2	699.00	1398.00
Yoga Mat	13	Priya Sharma	1	699.00	699.00

12. Number of times each product was ordered

product_id	product_name	times_ordered	total_quantity	total_revenue
7	Mixer Grinder 750W	2	2	6598.00
3	Non-stick Cookware Set	2	2	3798.00
6	Wireless Headphones	2	2	2998.00
17	Steel Water Bottle 1L	1	3	1197.00
4	Wings of Fire (Book)	1	3	750.00
2	Men Cotton T-Shirt	1	2	998.00
10	Organic Honey 500g	1	2	700.00
13	Power Bank 20000mAh	1	2	2598.00
15	Women Kurti	1	2	1398.00
18	Atomic Habits (Book)	1	1	499.00
11	Bluetooth Speaker	1	1	1799.00
9	Building Blocks Set	1	1	799.00
5	Kashmir Willow Cricket ...	1	1	1299.00
1	Redmi Note 13 5G	1	1	17999.00
16	Running Shoes	1	1	2299.00
12	Smart Watch	1	1	2499.00
14	USB-C Fast Charger 65W	1	1	999.00
19	Yoga Mat	1	1	699.00

13. Products never ordered

product_id	product_name
8	Face Wash 100ml
20	Moisturizer 200ml
22	Denim Jacket
23	Leather Wallet
24	Almonds 500g
25	Green Tea 100 Bags
26	Cotton Bedsheet
27	Cricket Ball Pack
NULL	NULL

14. Total quantity sold

total_quantity_sold
29

15. Customers with the maximum number of orders

customer_id	customer_name	total_orders
1	Arun Kumar	3

16. Total spending by each customer

customer_id	customer_name	total_orders	total_spent
1	Arun Kumar	3	22396.00
7	Vikram Singh	2	7297.00
3	Rahul Verma	2	7295.00
6	Divya Nair	2	6697.00
2	Priya Sharma	2	3393.00
5	Karthik Raja	1	1299.00
9	Suresh Babu	1	799.00
4	Sneha Reddy	1	750.00
8	Meena Iyer	0	0.00

17. Average order value

average_order_value
3566.14

18. Total sales amount

total_sales_amount
49926.00

19. Orders and sales by status

order_status	total_orders	total_amount
Pending	3	2998.00
Shipped	4	10242.00
Delivered	7	36686.00
Cancelled	1	398.00

Shree Pranava Ganesh N R

II BCA Batch 1